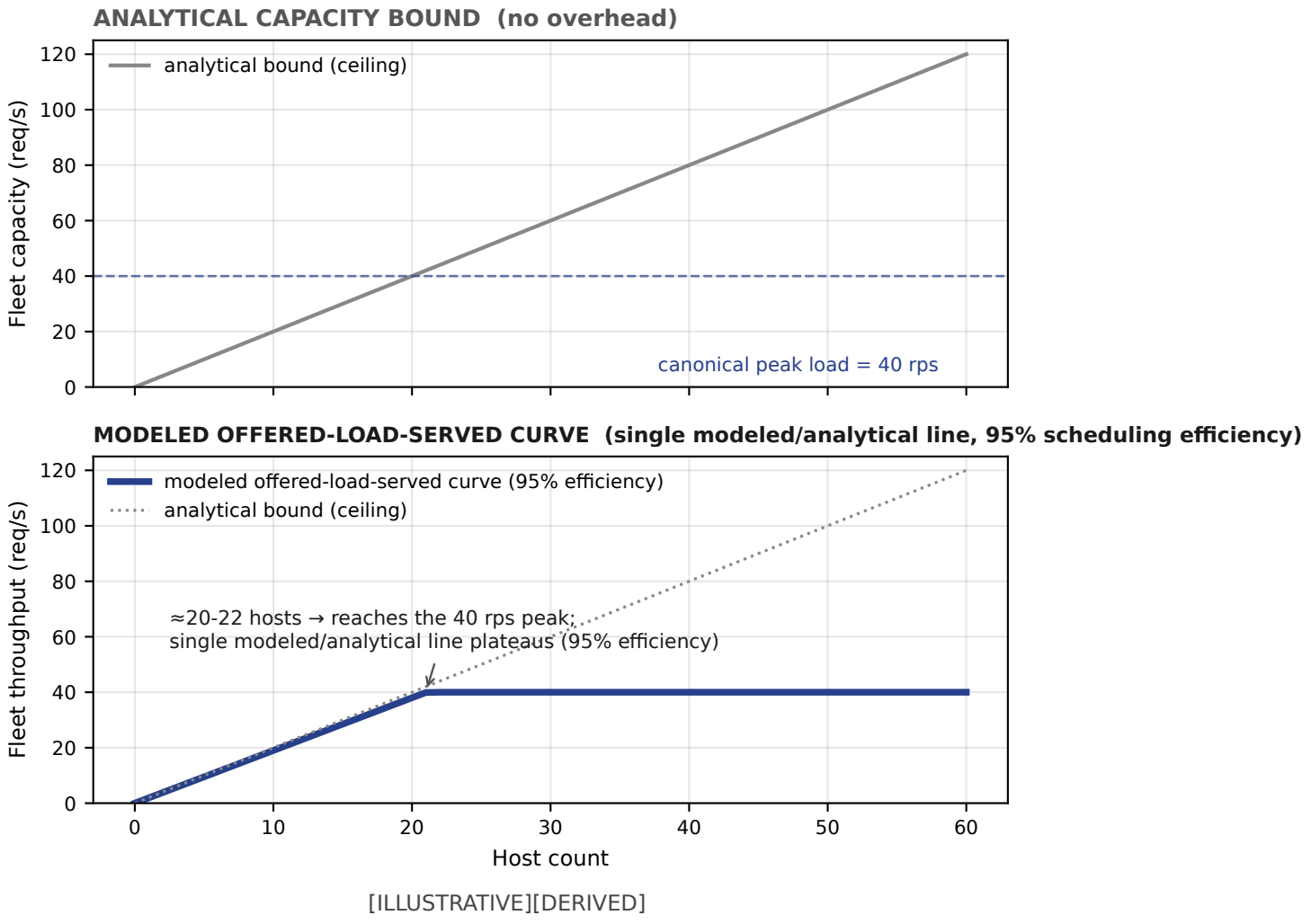
ANALYTICAL CAPACITY BOUND (no overhead)
analytical bound (ceiling)
Fleet capacity (req/s)
120
100
80
60
40
20
0
canonical peak load = 40 rps
MODELED OFFERED-LOAD-SERVED CURVE (single modeled/analytical line, 95% scheduling efficiency)
modeled offered-load-served curve (95% efficiency)
analytical bound (ceiling)
Fleet throughput (req/s)
≈20-22 hosts → reaches the 40 rps peak;
single modeled/analytical line plateaus (95% efficiency)
0
10
20
30
40
50
60
Host count
[ILLUSTRATIVE][DERIVED]